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MAGNETIC FLUID ASSIST DEVICES

Abstract

A magnetic fluid assist device can include a reservoir adapted to retain a fluid and a magnetic surface coupler attached to the reservoir. The magnetic surface coupler can magnetically connect to a surface supporting a target item, such as a seized joint. The surface coupler can also include a gasket configured to provide a fluid-sealed interface between the surface coupler and a target item. The device can also be configured for applying fluid to an inverted surface by including a rod extending along an interior central axis of the device, from an end of the reservoir opposite the target item, toward the target item. The rod can include a magnet affixed to an end corresponding to the target item and an adjustable nut configured for modifying an axial position of the magnet. The nut can be configured to enable control of a distance between the magnet and the target item.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation in part of U.S. Nonprovisional patent application Ser. No. 18/970,216, filed Dec. 5, 2024 which claims priority to U.S. Provisional Patent Application No. 63/606,303, filed Dec. 5, 2023, both of which are incorporated herein by reference.

BACKGROUND

[0002] Penetrating fluids, commonly known as penetrating oils or rust dissolvers, are widely used for loosening rusted or seized metal parts, facilitating the disassembly of mechanical components, and providing lubrication to enhance the performance of machinery. Traditional application methods, however, pose challenges when dealing with vertical or inverted surfaces, as the fluid tends to drip or run off, reducing the effectiveness of the technique.

[0003] Typical solutions to this problem include spray applicators and gel formulations. However, these solutions are still subject to gravity moving the fluid from where it is needed, which can cause increased costs from waste and lead to contamination of the environment around the target area.

SUMMARY

[0004] This document describes a magnetic fluid assist device. The magnetic fluid assist device can include a reservoir housing adapted to retain a volume of fluid and a magnetic surface coupler attached to the reservoir housing. The magnetic surface coupler can magnetically seal against a surface that is supporting a target item, such as a seized joint or fastener.

[0005] In some examples, the magnetic fluid assist device can also include a common interface. For example, the reservoir housing can include a portion of a common interface and the magnetic surface coupler can include another portion of the common interface. Further, the portion of the common interface and the other portion of the common interface can be configured to fit together to provide a fluid-sealed connection between the reservoir housing and the magnetic surface coupler, the fluid-sealed connection enabling fluid communication between the reservoir housing and the magnetic surface coupler.

[0006] Additionally, the common interface can further include a sealing element. The sealing element can be integrated with the portion of the common interface, integrated with the other portion of the common interface, or be a separate element.

[0007] In additional examples, the magnetic surface coupler can include at least one magnetic element. The at least one magnetic element can include a magnetic ring or at least one magnet oriented about a perimeter of the magnetic surface coupler. Further, in some examples, the magnetic surface coupler can include a gasket configured to provide a sealed interface between the magnetic surface coupler and a target item for the fluid. Additionally, the reservoir housing can be made from a flexible material that allows the reservoir housing to be fitted around obstacles while maintaining the sealed interface between the magnetic surface coupler and the target item.

[0008] In further examples, the gasket that is configured to provide the sealed interface is affixed to an interface surface of the magnetic surface coupler. The interface surface can be configured as a substantially planar interface surface or a non-planar interface surface. Further, in some examples, a cross-sectional shape of the interface surface comprises at least one of: a circle, an ellipse, a rectangle, a triangle, or a combination of one or more of the above.

[0009] This document also describes a device for applying fluid to an inverted surface. The device for applying fluid to an inverted surface can include a tube with an open end, one or more magnets disposed around the open end, a gasket surrounding the open end for sealing against a surface, an

opening at another end of the tube, opposite the open end, the opening configured to enable the addition of fluids into the tube, and a removable cap for sealing the opening. In some examples, the one or more magnets can be configured to secure the device to metal surfaces and seal the device to metal surfaces, via the gasket.

[0010] In some examples, the device for applying fluid to an inverted surface can also include a reservoir attached to the other end for holding fluids, the reservoir in fluid communication with the open end of the tube, via the opening at the other end of the tube. In some cases, the reservoir can be fitted with a removable cap for sealing the reservoir.

[0011] In additional examples, the device for applying fluid to an inverted surface can also include a rod extending along an interior central axis of the device, from a distal end of the reservoir opposite the other end of the tube, toward the open end of the tube, the rod penetrating the distal end of the reservoir via a fluid-sealed port. The device for applying fluid to an inverted surface can also include a central magnet affixed to an end of the rod that corresponds to the open end of the tube and an adjustable nut configured for modifying an axial position of the magnet. The adjustable nut can be configured to enable control of a distance between the central magnet and the surface to provide a fluid-sealed interface between the device and the surface and locking the central magnet in position. In some additional examples, the rod can also include a plunger configured to push fluid from the distal end of the reservoir toward the open end of the tube, effective to enable the fluid to maintain contact with the surface.

[0012] This document also describes a method for applying fluid to a target item. The method can comprise of connecting a reservoir of fluid to the target item using one or more magnets. The method can also include securing the reservoir to the target item while the reservoir is empty and adding penetrating oil fluid through a closeable opening. In some examples, the method can also include adjusting a position of the reservoir on the target item by manipulating the magnets.

[0013] In some examples, the reservoir includes a rod extending along an interior central axis of the reservoir, and at least one magnet attached to an end of the rod corresponding to an end of the reservoir configured for attachment to the target item, the rod penetrating the reservoir via a fluid-sealed port. The reservoir can also include an adjustable nut configured for modifying an axial position of the magnet. In such configurations, the method can also include securing the reservoir to the target item using the magnet on the rod, adjusting a distance between the magnet and the target item using the adjustable nut, and securing the magnet at the adjusted distance using the adjustable nut.

[0014] In additional examples, the rod also includes a plunger configured to push fluid toward the end of the reservoir configured for attachment to the target item. In such cases, the method can further comprise adjusting a distance between the plunger and the end of the reservoir configured for attachment to the target item, effectively to enable the fluid to maintain contact with the target item.

[0015] In other additional examples, the reservoir comprises a flexible reservoir, and the method can further comprise connecting the flexible reservoir to the target item using the magnets, conforming the flexible reservoir to an environment around the target item, and adding fluid to the flexible reservoir.

[0016] In this way, the magnetic fluid assist device can be used to securely hold a volume of a fluid, such as penetrating oil or fluid, in place on various surfaces, including horizontal, slanted, vertical, and inverted surfaces, to increase contact time and effectiveness.

[0017] There has thus been outlined, rather broadly, the more important features of the invention so that the detailed description thereof that follows may be better understood, and so that the present contribution to the art may be better appreciated. Other features of the present invention will become clearer from the following detailed description of the invention, taken with the accompanying drawings and claims, or may be learned by the practice of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1A illustrates an example implementation of a magnetic fluid assist device, which is suitable for use on vertical, horizontal, or slanted surfaces, according to some embodiments of the present disclosure.

[0019] FIG. 1B illustrates example cross-sectional views of an interface surface included with the magnetic fluid assist device of FIG. 1A, according to some embodiments of the present disclosure.

[0020] FIG. 1C illustrates side-views of example interface surfaces, each with a different planarity configuration, according to some embodiments of the present disclosure.

[0021] FIG. 1D illustrates an example magnetic fluid assist device that includes a flexible reservoir housing, according to some embodiments of the present disclosure.

[0022] FIG. 2A illustrates another example implementation of a magnetic penetrating fluid assist device, which is suitable for use on inverted surfaces, as well as vertical, horizontal, or slanted surfaces, according to some embodiments of the present disclosure.

[0023] FIG. 2B illustrates an example implementation of a magnetic fluid assist device which includes a plunger delivery mechanism for the penetrating fluid, which is suitable for use on vertical, horizontal, or slanted surfaces, according to some embodiments of the present disclosure.

[0024] FIG. 3 is a flowchart illustrating an example method of applying fluid to target items, according to some embodiments of the present disclosure.

[0025] These drawings are provided to illustrate various aspects of the invention and are not intended to be limiting of the scope in terms of dimensions, materials, configurations, arrangements or proportions unless otherwise limited by the claims.

DETAILED DESCRIPTION

[0026] While these example embodiments are described in sufficient detail to enable those skilled in the art to practice the invention, it should be understood that other embodiments may be realized and that various changes to the invention may be made without departing from the spirit and scope of the present invention. Thus, the following more detailed description of the embodiments of the present invention is not intended to limit the scope of the invention, as claimed, but is presented for purposes of illustration only and not limitation to describe the features and characteristics of the present invention, to set forth the best mode of operation of the invention, and to sufficiently enable one skilled in the art to practice the invention. Accordingly, the scope of the present invention is to be defined solely by the appended claims.

Definitions

[0027] In describing and claiming the present invention, the following terminology will be used.

[0028] The singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a magnet” includes reference to one or more of such materials and reference to “the interface” refers to one or more of such features.

[0029] As used herein with respect to an identified property or circumstance, “substantially” refers to a degree of deviation that is sufficiently small so as to not measurably detract from the identified property or circumstance. The exact degree of deviation allowable may in some cases depend on the specific context.

[0030] As used herein, “adjacent” refers to the proximity of two structures or elements.

Particularly, elements that are identified as being “adjacent” may be either abutting or connected. Such elements may also be near or close to each other without necessarily contacting each other. The exact degree of proximity may in some cases depend on the specific context.

[0031] As used herein, the term “about” is used to provide flexibility and imprecision associated with a given term, metric or value. The degree of flexibility for a particular variable can be readily determined by one skilled in the art. However, unless otherwise enunciated, the term “about”

generally connotes flexibility of less than 2%, and most often less than 1%, and in some cases less than 0.01%.

[0032] As used herein, a plurality of items, structural elements, compositional elements, and/or materials may be presented in a common list for convenience. However, these lists should be construed as though each member of the list is individually identified as a separate and unique member. Thus, no individual member of such list should be construed as a de facto equivalent of any other member of the same list solely based on their presentation in a common group without indications to the contrary.

[0033] As used herein, the term “at least one of” is intended to be synonymous with “one or more of.” For example, “at least one of A, B and C” explicitly includes only A, only B, only C, or combinations of each.

[0034] Any steps recited in any method or process claims may be executed in any order and are not limited to the order presented in the claims. Means-plus-function or step-plus-function limitations will only be employed where for a specific claim limitation all of the following conditions are present in that limitation: a) “means for” or “step for” is expressly recited; and b) a corresponding function is expressly recited. The structure, material or acts that support the means-plus function are expressly recited in the description herein. Accordingly, the scope of the invention should be determined solely by the appended claims and their legal equivalents, rather than by the descriptions and examples given herein.

EXAMPLE EMBODIMENTS

[0035] A technology is described for applying penetrating fluid to surfaces, and, in various example implementations, for a device designed to enhance the effectiveness of fluids, such as rust dissolvers, penetrating oils or gels, or other fluids, on vertical, slanted, horizontal, or inverted surfaces in various configurations. An initial overview of technology embodiments is provided below and then specific technology embodiments are described in further detail later. This initial summary is intended to aid readers in understanding the technology more quickly but is not intended to identify key features or essential features of the technology nor is it intended to limit the scope of the claimed subject matter.

[0036] In many environments, mechanical assemblies can become stuck or seized from rust, contaminants, or other causes. When the assembly needs maintenance, replacement, or other disassembly, fluids, such as penetrating fluids (and/or oils) and rust dissolvers are often used to loosen the seized parts. Typical application methods, however, can be less effective on vertical or inverted surfaces, as the fluid tends to drip or run off. The disclosed technology uses magnets to secure and seal a reservoir of penetrating fluid to an affected surface, which can reduce the amount of fluid needed and allow longer contact between the surface and the fluid. This can reduce costs by reducing the amount of needed fluid and the amount of cleanup necessary to address fluid that runs off the treated surface.

[0037] FIG. 1A illustrates an example implementation of a magnetic fluid assist device **100**, which is suitable for use on vertical, horizontal, or slanted surfaces, according to some embodiments of the present disclosure. The example implementation includes a reservoir housing **102** adapted to retain a volume of fluid and a magnetic surface coupler **104** attached to the reservoir housing. In example implementations, the fluid can include a rust dissolver, a penetrating oil or gel, or a combination of such fluids.

[0038] In some cases, the magnetic surface coupler **104** and the reservoir housing **102** can be integrated as a single unit (see FIG. 2B). However, in other cases, the magnetic surface coupler **104** and the reservoir housing **102** can be modular to allow for replaceable coupling for different sizes. The reservoir housing **102** can include a portion **106-1** of a common interface **106**. The magnetic surface coupler **104** can include another portion **106-2** of the common interface **106**. In some implementations, the portion **106-1** of the common interface and the other portion **106-2** of the common interface can be configured to fit together to provide a fluid-scaled connection between

the reservoir housing **102** and the magnetic surface coupler **104**. The fluid-sealed connection can enable fluid communication between the reservoir housing **102** and the magnetic surface coupler **104**. In this way, the magnetic fluid assist device **100** can be used on horizontal, vertical, or slanted surfaces to increase the effectiveness of rust dissolvers, penetrating fluids, gels, and oils, and reduce waste and contamination from dripping fluid.

[0039] The common interface **106** can be realized as any of a variety of suitable interfaces that can be sealed and allow fluid to flow between the reservoir housing **102** and the magnetic surface coupler **104**, such as threaded interface or a press-fit interface. As shown in FIG. **1A**, the portion **106-1** is a plug and the other portion **106-2** is a receptacle that can receive the plug. In other implementations, the locations plug and receptacle may be switched between the reservoir housing **102** and the magnetic surface coupler **104**. In some implementations, the common interface **106** also includes a sealing element **108**. The sealing element **108** can be integrated with the portion **106-1** of the common interface (as shown in FIG. **1A**), integrated with the other portion **106-2** of the common interface **106**, or be a separate element. The sealing element **108** can be made from any suitable material and realized as any of a variety of sealing devices, including a gasket or an O-ring.

[0040] In some examples, the magnetic surface coupler **104** includes at least one magnetic element **110**. The at least one magnetic element **110** can comprise a magnetic ring or at least one magnet oriented about a perimeter of the magnetic surface coupler **104**. As shown in FIG. **1A**, the magnetic surface coupler **104** includes four magnetic elements **110**, spaced around the perimeter of the magnetic surface coupler **104** (for clarity, only three of the four magnetic elements **110** are shown in FIG. **1A**). In other implementations, a different quantity or arrangement of magnetic elements **110** may be used.

[0041] In additional examples, the magnetic surface coupler **104** can include a gasket **112** that is configured to provide a sealed interface between the magnetic surface coupler **104** and a target item for the fluid. In other examples, the gasket may be replaced with another sealing element, such as an O-ring. In additional examples, the gasket that is configured to provide the sealed interface is affixed to an interface surface **114** of the magnetic surface coupler.

[0042] A cross-sectional shape of the interface surface **114** can be any of a variety of shapes, including, by way of non-limiting example, a circle, an ellipse, a rectangle, a triangle, or a combination of one or more of the above. Further, the interface surface **114** can be configured as a substantially planar interface surface or a non-planar interface surface, including by way of non-limiting example, substantially planar and non-planar. For example, FIG. **1B** illustrates four example cross-sectional views **116** (**116-1**, **116-2**, **116-3**, and **116-4**), each with different cross-sectional shapes (e.g., circular, elliptical, rectangular, and triangular). Further, any shape can also be made in different sizes, as appropriate to the shape of the target item. Similarly, FIG. **1C** illustrates side-views of three example interface surfaces **114** (**114-1**, **114-2**, and **114-3**), each with different planarity (e.g., substantially planar, stepped, and slanted). It is noted that other planarity configurations (not shown in FIG. **1C**) can also be used with the example magnetic fluid assist device **100**.

[0043] Returning to FIG. **1A**, in some implementations, the reservoir housing **102** can also include a removable cap **118** to retain fluids in the reservoir housing **102** after filling. Although not shown on FIG. **1A**, the removable cap **118** may be secured to the reservoir housing **102** by various means (e.g., a leash or captive screw mechanism) to reduce the likelihood of losing the removable cap **118**.

[0044] In additional examples, the reservoir housing **102** can be made from any of a variety of suitable materials, including, for example, plastics, rubbers, metals, and silicone. In additional implementations, the reservoir housing **102** can be made from flexible material that allows a user to conform the reservoir around obstacles while maintaining the sealed interface between the magnetic surface coupler and the target item. For example, FIG. **1D** illustrates an example

magnetic fluid assist device **100-1** that includes a flexible reservoir housing **102-1** and a magnetic surface coupler **104** (as described with reference to FIGS. **1A** through **1C**). The flexible reservoir housing **102-1** is configured as a tray for holding fluid and includes a removable cap **118-1** to hold the fluid within the flexible reservoir housing **102-1**. The view in FIG. **1D** is a top view of the flexible reservoir housing **102-1** and the removable cap **118-1**. As shown, the example magnetic fluid assist device **100-1** has been flexed around an obstructing bolt-head **120**. In this case, the bolt-head **120** is not a target of loosening but is rather an obstruction to the reservoir. Of course, other obstructions can be avoided with such a flexible reservoir such as, but not limited to, flanges, protrusions, struts, supports, and the like. The unflexed position of the flexible reservoir housing **102-1** is shown in dashed lines.

[0045] FIG. **2A** illustrates another example implementation of a magnetic fluid assist device **200**, which is suitable for use on inverted surfaces, as well as vertical, horizontal, or slanted surfaces, according to some embodiments of the present disclosure. The example implementation includes a tube **202** with an open end **204** and one or more magnets **206** disposed around the open end **204**. The example implementation also includes a gasket **208** surrounding the open end **204** for sealing against a surface and an opening **210** at another end of the tube opposite the open end **204**. In some examples, the gasket **208** can be configured as an O-ring. The opening **210** can be configured to enable the addition of fluids into the tube **202**. In some implementations, the example device also includes a removable cap **212** for sealing the opening **210**.

[0046] In some example implementations, the magnets **206** can be configured to secure the device **200** to metal surfaces and seal the device **200** to metal surfaces, via the gasket **208**. In some additional examples, the magnets **206** can be a magnetic ring or at least one magnet oriented about a perimeter of the tube **202**. As shown in FIG. **2A**, the one or more magnets **206** include four magnetic elements, spaced around the perimeter of the tube **202** (for clarity, only three of the four magnets are shown in FIG. **2A**). In other implementations, a different quantity or arrangement of magnets may be used.

[0047] In some examples, the example device **200** also includes a reservoir **214** attached to the other end opposite the open end **204**. The reservoir **214** can be configured for holding fluids. Further, the reservoir **214** is in fluid communication with the open end **204** of the tube **202**, via the opening **210**. In some implementations, the reservoir **214** can be fitted with a removable cap **216** for sealing the reservoir (for clarity, the removable cap **216** is shown with dashed lines). As noted above, either or both of the removable caps **212** and **216** may be secured to the tube **202** and the reservoir **214**, respectively, by various means (e.g., a leash or captive screw mechanism) to reduce the likelihood of losing either or both of the removable caps **212** and **216**.

[0048] In additional examples, the tube **202** can include a portion **218-1** of a common interface **218**. The reservoir **214** can include another portion **218-2** of the common interface **218**. In some implementations, the portion **218-1** and the other portion **218-2** can be configured to fit together to provide a fluid-sealed connection between the tube **202** and the reservoir **214**. The fluid-sealed connection can enable fluid communication between the tube **202** and the reservoir **214**. In this way, the example magnetic fluid assist device **200** can be used on inverted, horizontal, vertical, or slanted surfaces to increase the effectiveness of rust dissolvers, penetrating fluids, gels, and oils, and reduce waste and contamination from dripping fluid.

[0049] The common interface **218** can be realized as any of a variety of suitable interfaces that can be sealed and allow fluid to flow between the tube **202** and the reservoir **214**, such as threaded interface or a press-fit interface. As shown in FIG. **2A**, the portion **218-2** is a plug and the other portion **218-1** is a receptacle that can receive the plug. In other implementations, the locations plug and receptacle may be switched between the tube **202** and the reservoir **214**. In some implementations, the common interface **218** also includes a sealing element **220**. The sealing element **220** can be integrated with the other portion **218-2** of the common interface **218** (as shown in FIG. **2A**), integrated with the portion **218-1** of the common interface **218**, or be a separate

element. The sealing element **220** can be made from any suitable material and realized as any of a variety of sealing devices, including a gasket or an O-ring.

[0050] In some example implementations, the device **200** can also include a rod **222** that extends along an interior central axis of the device **200**, from a distal end of the reservoir **214** opposite the other end **210** of the tube, toward the open end **204** of the tube. In some cases, the rod **222** penetrates the distal end of the reservoir **214** via a fluid-sealed port **224**. The device **200** can also include a central magnet **226** affixed to an end of the rod **222** that corresponds to the open end **204** of the tube **202**.

[0051] The device **200** can also include an adjustable nut **228** configured for modifying an axial position of the magnet. The adjustable nut **228** can be configured to enable control of a distance between the central magnet **226** and the surface to provide a fluid-sealed interface between the device and the surface. The adjustable nut **228** can also enable locking the central magnet **226** in position. In this way, the example device **200** (with fluid inside) can be attached and secured to an inverted surface with the central magnet **226** (and or the magnets **206**) and create a seal against the surface. Notably, the central magnet **226** can be used alone as the magnetic element, or can include periphery magnets **206**. In some additional examples, the rod **222** can also include a plunger (not shown in FIG. 2A) that is configured to push fluid from the distal end of the reservoir **214** toward the open end **204** of the tube **202**. The plunger can thereby be effective to enable the fluid to maintain contact with the surface. In implementations that include the plunger, the tube **202** can also include a pressure-release mechanism to allow air in the headspace to escape as the fluid level rises (not shown in FIG. 2A). For example, one or more one-way valves oriented around a circumference of the reservoir **214**, near the other end **210** of the tube **202**.

[0052] FIG. 2B illustrates another variation of the magnetic fluid assist device **250** which includes a plunger delivery mechanism **252** for the penetrating fluid. In this example, the magnetic fluid assist device **250** includes a reservoir housing **254** which is configured to retain a penetrating fluid within the reservoir **256**. A magnetic surface coupler **258** can include one or more magnetic elements **259**. The magnetic elements can include a single ring magnet (as illustrated) or multiple magnetic elements (see FIG. 1A) oriented around a periphery of a contact opening. As with the other examples described herein, the contact opening can allow fluid to pass from the reservoir to a corresponding surface which includes a fastener to be loosened. An optional gasket **260** can be used to create a seal between the reservoir housing **254** and the surface (not shown) at an interface surface **262**.

[0053] The plunger delivery mechanism **252** can include a flexible hose **264** which is fluidly connected between a plunger outlet **266** and a reservoir cap **268**. The reservoir cap **268** can be engaged via a threaded connection or other suitable securing mechanism (e.g. interference, detents, etc). The plunger delivery mechanism **252** can include a main fluid reservoir body **270** and a sliding piston **272** which is actuated via a handle **274** to force fluid from inside the reservoir body **270** into the flexible hose **264** and into the reservoir **256** for contact with the fastener. In one option, an outlet end **276** of the flexible hose **264** can be inserted into the reservoir **256** to allow withdrawal of the penetrating fluid.

[0054] In addition to the example magnetic fluid assist devices described above, the present disclosure also describes methods of using the magnetic fluid assist devices. FIG. 3 is a flowchart illustrating an example method **300** of applying fluid to target items, according to some embodiments of the present disclosure. The method **300** can be implemented using some or all of the magnetic fluid assist devices, features, and techniques described above with reference to FIG. 1A through FIG. 2B.

[0055] The method includes, at block **310**, connecting a reservoir configured to hold fluid to the target item using one or more magnets. In some cases, the connection of the reservoir can be achieved via any of the magnetic surface couplers **104** described with reference to FIG. 1A through FIG. 2. In example implementations, the fluid can include a rust dissolver, a penetrating oil or gel,

or a combination of such fluids.

[0056] Further, the example method **300** can include additional optional steps **320-340**. For example, at block **320**, the method further includes securing the reservoir to the target item while the reservoir is empty. At block **330**, the method also includes adding fluid through a closeable opening. The method can also include, at block **340**, adjusting a position of the reservoir on the surface by manipulating the magnets.

[0057] In some cases, the reservoir can be configured as a flexible reservoir. In those configurations, the method **300** can optionally include (not shown in FIG. 3) one or more of: connecting a flexible reservoir to the target item using the magnets, conforming the flexible reservoir to an environment around the target item, and adding fluid to the flexible reservoir.

[0058] Additionally, in some cases similar to those described above, the reservoir can include a rod extending along an interior central axis of the reservoir and at least one magnet attached to an end of the rod corresponding to an end of the reservoir configured for attachment to the target item. The rod can penetrate the reservoir via a fluid-sealed port. The reservoir can also include an adjustable nut configured for modifying an axial position of the magnet.

[0059] In those configurations, the method **300** can include additional optional steps (also not shown on FIG. 3). For example, the method can further include securing the reservoir to the target item using the magnet on the rod, adjusting a distance between the magnet and the target item using the adjustable nut, and securing the magnet at the adjusted distance using the adjustable nut.

[0060] In additional examples, the rod can also include a plunger configured to push fluid toward the end of the reservoir configured for attachment to the target item. In those configurations, the method can further comprise (also not shown on FIG. 3) adjusting a distance between the plunger and the end of the reservoir configured for attachment to the target item, effective to enable the fluid to maintain contact with the target item.

EXAMPLES

[0061] Three nuts were welded to a 1/8" piece of steel and mounted in a vice. Each fastener was dipped in a solution of iodized salt, white vinegar, and hydrogen peroxide then torqued to 40 ft/lbs using a click type torque wrench in order to accelerate rusting. The fasteners were left overnight to corrode. About 12 hours later the control fastener was treated with PB BLASTER penetrating spray on both the front and the back side of the fastener and left to soak for over 12 hours. The test fastener was covered with the penetrating fluid applicator (as illustrated in FIG. 2B), filled at least half full with PB BLASTER penetrating fluid, and left to soak for over 12 hours.

[0062] When removing the fasteners the same torque wrench was used on increasing torque settings until the fastener broke loose without activating the click action on the torque wrench. The distance the penetrating fluid traveled into the threads was also observed.

[0063] Six tests were conducted and recorded, resulting in an average 7% reduction of torque needed to break loose the test fastener treated with the penetrating fluid applicator. These results were compared to the first fastener that was not treated with penetrating fluid and showed there was not a reliable difference between the untreated fastener and the fastener that was only sprayed using the aerosol can. Further, there was a 75% reduction of wasted fluid per test when the excess from the invention test bolt was drained into a cup and reused. In addition, virtually no fluid contaminated the floor when drained properly.

[0064] While the flowcharts presented for this technology may imply a specific order of execution, the order of execution may differ from what is illustrated. For example, the order of two more blocks may be rearranged relative to the order shown. Further, two or more blocks shown in succession may be executed in parallel or with partial parallelization. In some configurations, one or more blocks shown in the flow chart may be omitted or skipped.

[0065] These examples, implementations, embodiments, and configurations collectively provide a comprehensive solution for applying fluids in diverse scenarios, offering improved efficiency and usability over existing methods of application. The solutions can also reduce waste and

contamination in the application environment. In view of the limitations of the related art, the present invention represents a significant advancement in at least the field of penetrating fluid application devices, providing a versatile and effective solution for treating surfaces in various orientations.

[0066] Reference was made to the examples illustrated in the drawings and specific language was used herein to describe the same. It will nevertheless be understood that no limitation of the scope of the technology is thereby intended. Alterations and further modifications of the features illustrated herein and additional applications of the examples as illustrated herein are to be considered within the scope of the description.

[0067] Furthermore, the described features, structures, or characteristics may be combined in any suitable manner in one or more examples. In the preceding description, numerous specific details were provided, such as examples of various configurations to provide a thorough understanding of examples of the described technology. It will be recognized, however, that the technology may be practiced without one or more of the specific details, or with other methods, components, devices, etc. In other instances, well-known structures or operations are not shown or described in detail to avoid obscuring aspects of the technology.

[0068] Although the subject matter has been described in language specific to structural features and/or operations, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features and operations described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims. Numerous modifications and alternative arrangements may be devised without departing from the spirit and scope of the described technology.

Claims

1. A magnetic fluid assist device comprising: a reservoir housing adapted to retain a volume of fluid; and a magnetic surface coupler attached to the reservoir housing.
2. The device of claim 1, wherein: the reservoir housing includes a portion of a common interface; the magnetic surface coupler includes another portion of the common interface; and the portion of the common interface and the another portion of the common interface are configured to fit together to provide a fluid-sealed connection between the reservoir housing and the magnetic surface coupler, the fluid-sealed connection enabling fluid communication between the reservoir housing and the magnetic surface coupler.
3. The device of claim 2, wherein: the common interface further comprises a sealing element; and the sealing element is: integrated with the portion of the common interface; integrated with the other portion of the common interface; or a separate element.
4. The device of claim 1, wherein the magnetic surface coupler includes at least one magnetic element.
5. The device of claim 4, wherein the at least one magnetic element comprises: a magnetic ring; or at least one magnet oriented about a perimeter of the magnetic surface coupler.
6. The device of claim 1, wherein the magnetic surface coupler includes a gasket configured to provide a sealed interface between the magnetic surface coupler and a target item for the fluid.
7. The device of claim 6, wherein the reservoir housing is made from a flexible material that allows the reservoir housing to be fitted around obstacles while maintaining the sealed interface between the magnetic surface coupler and the target item.
8. The device of claim 6 wherein: the gasket that is configured to provide the sealed interface is affixed to an interface surface of the magnetic surface coupler; and the interface surface is configured as: a substantially planar interface surface; or a non-planar interface surface, and wherein: a cross-sectional shape of the interface surface comprises at least one of: a circle, an ellipse, a rectangle, a triangle, or a combination thereof.

- 9.** A device for applying fluid to an inverted surface, comprising: a tube with an open end; one or more magnets disposed around the open end; a gasket surrounding the open end for sealing against a surface; an opening at another end of the tube, opposite the open end, the opening configured to enable addition of fluids into the tube; and a removable cap for sealing the opening.
- 10.** The device of claim 9, wherein the one or more magnets are configured to: secure the device to metal surfaces; and seal the device to metal surfaces, via the gasket.
- 11.** The device of claim 9, further comprising a reservoir attached to the another end for holding fluids, the reservoir in fluid communication with the open end of the tube, via the opening at the other end of the tube.
- 12.** The device of claim 11, wherein the reservoir is fitted with a removable cap for sealing the reservoir.
- 13.** The device of claim 11, further comprising: a rod extending along an interior central axis of the device, from a distal end of the reservoir opposite the another end of the tube, toward the open end of the tube, the rod penetrating the distal end of the reservoir via a fluid-sealed port; a central magnet affixed to an end of the rod that corresponds to the open end of the tube; and an adjustable nut configured for modifying an axial position of the magnet, the adjustable nut configured to enable: control of a distance between the central magnet and the surface to provide a fluid-sealed interface between the device and the surface; and locking the central magnet in position.
- 14.** The device of claim 13, wherein the rod also includes: a plunger configured to push fluid from the distal end of the reservoir toward the open end of the tube, effective to enable the fluid to maintain contact with the surface.
- 15.** A method for applying fluid to a target item, comprising: connecting a reservoir configured to hold fluid to the target item using one or more magnets.
- 16.** The method of claim 15, further comprising: securing the reservoir to the target item while the reservoir is empty; and adding fluid through a closeable opening.
- 17.** The method of claim 15, further comprising adjusting a position of the reservoir on the target item by manipulating the magnets.
- 18.** The method of claim 15, wherein: the reservoir includes: a rod extending along an interior central axis of the reservoir, and at least one magnet attached to an end of the rod corresponding to an end of the reservoir configured for attachment to the target item, the rod penetrating the reservoir via a fluid-sealed port; and an adjustable nut configured for modifying an axial position of the magnet, and wherein the method further comprises: securing the reservoir to the target item using the magnet on the rod; adjusting a distance between the magnet and the target item using the adjustable nut; and securing the magnet at the adjusted distance using the adjustable nut.
- 19.** The method of claim 18, wherein: the rod also includes a plunger configured to push fluid toward the end of the of the reservoir configured for attachment to the target item, and wherein the method further comprises: adjusting a distance between the plunger and the end of the reservoir configured for attachment to the target item, effective to enable the fluid to maintain contact with the target item.
- 20.** The method of claim 15, wherein: the reservoir comprises a flexible reservoir, and the method further comprises: connecting the flexible reservoir to the target item using the magnets; conforming the flexible reservoir to an environment around the target item; and adding fluid to the flexible reservoir.
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